



CSS exercises/learning

Introduction to CSS: Lessons & Exercises

If HTML is the **skeleton** of a webpage, CSS (Cascading Style Sheets) is the **skin, clothing, and style**. CSS is what we use to add colors, change fonts, control spacing, and arrange the layout of our website.

Please look at this slideshow before viewing this material on the notion page:
[HTML + CSS slides](#)

Part 1: Concepts to Learn

1. CSS Syntax (How it Looks)

CSS works by targeting an HTML element and applying styles to it. A CSS rule is made up of a **selector** and a **declaration block**.

- **Selector:** Points to the HTML element you want to style (e.g., `h1`, `p`, `ul`).
- **Property:** The feature you want to change (e.g., `color`, `font-size`, `background-color`).
- **Value:** The setting you want to apply (e.g., `red`, `20px`, `blue`).

CSS

```
/* Example rule */  
p {  
  color: blue;  
  font-size: 16px;  
}
```

2. The Box Model

Every element on an HTML page is treated as a rectangular box. To control space around elements, you need to understand the **Box Model**, which consists of four layers:

1. **Content:** The actual text or image inside the element.
2. **Padding:** Clear space *inside* the element, directly around the content (inside the border).
3. **Border:** A line that goes around the padding and content.
4. **Margin:** Clear space *outside* the border, used to separate this element from other elements on the page.

3. Selectors: Classes vs. IDs

While you can style raw tags like `h1`, you will eventually want to style specific elements differently. We use HTML attributes to do this:

- **Class (`.className`):** Used for styles that you want to reuse on **multiple** elements across the page. In CSS, you target a class using a dot (`.`).
- **ID (`#idName`):** Used to target a **single, unique** element on the page. In CSS, you target an ID using a hashtag (`#`).

HTML

```
<p class="error-text">This is a red error message.</p>  
<p id="featured-paragraph">This is a unique paragraph.</p>
```

CSS

```
/* CSS */
.error-text { color: red; }
#featured-paragraph { font-weight: bold; }
```

Part 2: Starter HTML & CSS Template

To practice CSS, we use the `<style>` tag inside the `<head>` of our HTML document. This is called **Internal CSS**. Copy this code and save it as `style_practice.html` to complete your exercises.

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Learning Page</title>

  <style>
    /* YOUR CSS CODE WILL GO HERE */

  </style>
</head>
<body>

  <h1 id="main-title">Welcome to CSS Styling</h1>

  <p class="intro-text">This is the introductory paragraph
that needs some style.</p>

  <p>This is a regular paragraph that should stay normal.</
p>
```

```
<div class="box-element">
    Inside the box!
</div>

</body>
</html>
```

Part 3: Coding Exercises

Before attempting these exercises, please make an account on CodVerter and copy and paste your project link into your pre-curriculum submission. Thank you!

[Login for CodVerter](#)

Difficulty 1: Easy (Selectors & Basic Styling)

Goal: Learn how to differentiate and target HTML elements using Class and ID attributes.

- **Task 1:** Specify a unique ID for a main title element using the `id` attribute, then choose the correct CSS selector symbol (`#`) to target that specific element and change its text color.
- **Task 2:** Assign a reusable class attribute to an introductory paragraph. Use the proper CSS class prefix (`.`) to target it, setting its `font-size` using pixel units (`px`) and making its `font-style` italic.

Suggested AI Prompt for Students:

"I am learning CSS selectors. I need to style a unique title element using its ID selector and an intro paragraph using its class selector. **Don't write the CSS rules for me.** Ask me for my initial code setup, and guide me on using the correct prefix symbols."

Difficulty 2: Medium (The Box Model)

Goal: Visualise and control spacing using margin, borders, and padding.

- **Task 1:** Target a block element with a class name of `.box-element` inside your stylesheet.
- **Task 2:** Apply the Box Model components to it by assigning a custom background color, a defined layout width in pixels, and a clearly visible `solid` border outline.
- **Task 3:** Create spacing *inside* the element between the content and the border using padding, and add a separate margin to create empty space *outside* the element to separate it from surrounding tags.

Suggested AI Prompt for Students:

"I need to configure the CSS Box Model on a class called `.box-element`. I want to define a width, background, a solid border, margin, and padding. **Do not give me the final code.** Ask me to draft the rules first, then check if I mixed up inner versus outer spacing."

Difficulty 3: Hard (Pseudo-classes & Hover States)

Goal: Apply state-based selectors and handle simple user interaction overrides.

- **Task 1:** Target the outermost `body` element of your webpage to apply a default background color across the entire viewable page area.
- **Task 2:** Implement a CSS rule using a pseudo-class selector attached to your `.box-element` class that only activates when a user hovers their mouse cursor directly over it.
- **Task 3:** Within that hover state rule, change both the background color and the text color to contrasting colors, forcing a visual feedback change when hovered.

Suggested AI Prompt for Students:

"I want to style the `body` element and append a `:hover` pseudo-class to a box element so its text and background swap colors on mouseover. **Don't give me the code.** Prompt me for my selector syntax and coach me through getting the hover rule working."